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Deployable Self-Referenced Features for Cross-Cell Battery State-of-Health Estimation: Randomized Pairing Controls and Benchmarking

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Abstract

Battery health estimation depends on feature references as well as learners. Changing one column's normalization moves root-mean-square error (RMSE) by up to 108%. We construct a self-referenced lift (SRL) from each cell's earliest retained record. Under the offline author protocol, learners with the full history stack improve on 9 of 11 units after Holm correction, with none significantly worse and two units per backbone not significantly different. Relative-change medians over all 11 units are −16.8% for the frozen foundation model and −19.1% for gradient boosting, including learner effects. Under a strict online reader excluding target-future and label information, compact SRL instead yields +4.3% for gradient boosting (5 of 11 improve) and +10.4% for the frozen foundation model (3 of 11 improve): no overall online accuracy gain. Legacy offline pairing controls favour self-pairing ($p = 0.0099$, 100 derangements). A companion benchmark with implicit self-comparison improves in 6 of 6 batches with per-batch selection (5 of 6 with a fixed configuration); an unseen-library legacy diagnostic improves by 22–33%. Deployability concerns feature computability, not a strict-online accuracy advantage.

Keywords: battery state of health; lithium-ion batteries; feature engineering; self-referenced representation; tabular foundation model; battery management system

1. Introduction

Battery energy storage is central to sustainable power systems: it absorbs renewable variability, provides frequency and capacity services, and extends the life of costly, carbon-intensive assets. All these functions depend on knowing how healthy each cell is. State-of-health (SOH) estimation is thus not merely a diagnostic convenience but an operational and economic requirement for battery management systems (BMS) in grid-scale storage and electrified transport: accurate estimates enable deeper usable state-of-charge windows, better-timed maintenance, and defensible second-life decisions, reducing lifetime cost and material footprint.

Data-driven SOH estimation has advanced rapidly: a prominent line embeds empirical degradation equations into neural networks as soft constraints, establishing the four processed libraries, official cell-level splits, and per-experiment results used here [1], while a companion benchmark supplies five deep baselines under three input types [2]. It has since grown through 2025–2026 with physics-informed variants [3–5], physics-embedded transfer learning [6], few-shot cross-domain frameworks [7], and quantum-feature hybrids [8]. A

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smaller parallel literature studies features, not architectures: a single interpretable quantity from early-cycle discharge curves predicts cycle life with a linear model [9]; cycle-difference features populate open battery machine-learning platforms [10]; recent work pairs interpretable feature fragments with spatio-temporal learners [11]; and in the wider time-series and tabular literature, simple transforms with standard learners form an established genre [12,13]. The closest of these, the early-cycle discharge-curve difference of [9], is also a within-cell comparison: a single hand-designed quantity, read from a fixed early window, predicting end of life. Here, however, the reference itself is the object of study: which past state each cycle is compared against, whether the pairing is responsible for the accuracy it buys, and whether it survives being computable in service. A third development is tabular foundation models, which infer training-free on small tabular tasks [14] yet require explicit time-derived features for temporal problems [15]; two recent efforts bring them into battery health—one distilling a time-series model [16], one coupling a tabular model to a physics-informed network by residual correction [17], and both are training-based hybrids—while large multi-source battery models begin to appear [18]. Retrieval-based context construction for these models works on i.i.d. benchmarks [19,20].

Nearly all of this work rests on a rarely examined assumption: the representation fed to the model—per-cycle statistical features under standard normalization—is settled preprocessing. Two measurements contradict it: first, with data, model, split and every other column fixed and only the cycle-index normalization convention changed, pooled-cycle RMSE moves by $+107.8\%$ on one evaluation unit, -49.6% on another, and -0.5% on a third; second, the benefit of an explicit self-referenced feature on the 67-dimensional feature set reverses sign, from -5.3% to $+4.5\%$, depending solely on whether features are normalized per cell or against source-fitted statistics. Normalization is where the representation's *comparison target* is decided, and it materially affects accuracy under the tested protocol.

The motivation is a feature-reference question, not an identity imposed by the label: the benchmark SOH is present discharge capacity divided by nominal capacity, not the measured first-cycle capacity. Comparing each unit against its own baseline is standard practice in other longitudinal settings: change-from-baseline and ANCOVA analyses in clinical trials [21], within (fixed-effect) transformations in panel econometrics [22], and explicit handling of unknown initial wear in prognostics datasets [23]. Battery pipelines are implicitly consistent with this: normalizing each cell by its own full-lifetime extrema compares every cell against itself. That convention suits offline assessment with complete trajectories, but cannot be reproduced in service, because cells differ widely in lifetime and a cell in operation does not know its own lifetime extrema. Total life is not the cycle-level SOH target studied here; nevertheless, the offline-helpful comparison is unavailable online.

This paper makes the comparison explicit, minimal, and deployment-computable, then tests—with a randomized mispairing control rather than a causal identification argument—whether the pairing itself is responsible for the observed gains. Our contributions are as follows.

1. We identify and quantify *what features are compared against* as a first-order design variable in degradation feature engineering, hidden inside routine normalization choices, and show its effect spans -49.6% to $+107.8\%$ across eleven evaluation units.
2. We define the self-referenced lift (SRL), an explicit, deployment-computable, model-agnostic feature family, and separate its computability from its accuracy. The offline author protocol yields Holm-significant improvements over a physics-informed network on 9 of 11 units for each of two backbones, including learner effects; the corrected strict online evaluation does not reproduce an overall accuracy gain from adding compact SRL.

3. We introduce a randomization-based validation protocol for representation claims—randomized mispairing, affine negative controls, permutation inference, and preregistration [24–27]—and use it to show that self-pairing, not generic trend information, is the active ingredient. Because the control manipulates the feature matrix and not the ageing process, this is evidence about the representation, not an interventional claim about the cell (Section 2.6).
4. We describe a model- and protocol-specific carrier pattern consistent with HUST and TJU-3, with counterexamples on other units, and give the operating conditions for the method. The attribution is measured for the tree model under the author-protocol normalization; the foundation-model probe of Section 3.10 lacks that structure, so we present it as a model- and protocol-scoped account, not as a property of the data.

The paper proceeds as follows. Section 2 describes the datasets, feature families, two evaluation pipelines, models, and statistical protocol; Section 3 reports the results; Section 4 discusses the mechanism, comparability with published numbers, battery-management implications, and limitations; Section 5 concludes.

2. Materials and Methods

For readers coming from an electrochemistry background, the machine-learning vocabulary used below is fixed as follows. *Cross-cell estimation* means that the model is fitted on cells whose full trajectories are available and tested on cells it has never seen; the latter are the *held-out cells* of an official split, reserved by the dataset authors. Primary-benchmark settings are not selected on them; the companion best-of-four summary is explicitly a post hoc choice using held-out scores and is separated from its fixed configuration (Section 3.5). An *evaluation unit* is the group of cells over which results are averaged, here one ageing batch or one library. A *feature family* is a set of columns obtained from one and the same comparison; an *ablation* removes one family at a time to measure what it contributes. *Permutation importance* measures how much error rises when a column's values are shuffled, and a *permutation test* asks whether an observed difference survives relabelling. A *derangement* is a relabelling with no element left in place, and a *negative control* is an arm whose true effect is known to be zero, so that it can only reveal artefacts of the measurement itself.

2.1. Datasets and Splits

The primary benchmark consists of the four processed libraries released with the reference physics-informed work [1]: XJTU (six ageing batches), TJU (three batches) [28], HUST [29], and MIT [9], giving 11 evaluation units in total (387 cells, 82 of them held out); the MIT unit pools the three original sub-batches of that release (12 May 2017, 30 June 2017 and 12 April 2018) into a single unit, so it is not broken out per sub-batch as the XJTU batches are. The libraries span three cathode chemistries (NCM, NCA and LFP), with one batch that mixes two of them, and nominal capacities from 1.1 to 3.5 Ah; Appendix A lists chemistry, nominal capacity, cell counts, per-cell cycle-count ranges and the lowest observed SOH for every unit. We adopt the official cell-level splits without modification; held-out cells are never used for fitting or model selection; normalization statistics follow the pipeline and are stated per pipeline below. In particular, Pipeline A scales each cell by its own full-lifetime extrema, which is same-cell future information that a cell in service cannot know; we treat that convention as offline-only and never pool its numbers with Pipeline B. The secondary benchmark is the companion deep-learning benchmark [2] on XJTU (55 cells, 67 handcrafted features, leave-one-cell-out splits). For external verification, we use an independent library, MATR-CLO (45 cells), the closed-loop-optimisation batch of Attia et al. [30]. It is disjoint from the three MIT batches in the primary benchmark and from the MATR batch used during development, and it took part in no design or

selection decision; every use of it was preceded by criteria frozen in the corresponding experiment script.

Cell counts per evaluation unit are fixed by the official protocol and range from two or three held-out cells (each XJTU batch, TJU-3) to 20–23 held-out cells (HUST, MIT); this constrains per-unit statistical power and is treated explicitly in Section 4.5.

2.2. Feature Families and the Self-Referenced Lift

Each cycle t of a cell is described by a 16-dimensional vector $\mathbf{x}_t$ of charge-segment statistics: the mean, standard deviation, skewness, kurtosis, slope and entropy of the voltage and current series, plus the constant-current and constant-voltage capacities and their durations (the columns CC Q, CC charge time, CV Q and CV charge time of the release), as released with the reference protocol [1]. Candidate feature lifts differ in *what each cycle is compared against*; Table 1 classifies them.

Table 1. Candidate feature families classified by comparison target. "Online" indicates whether the quantity can be computed during service, when the cell's total life is unknown.

Family	Definition	Comparison Target	Class	Online
raw	$\mathbf{x}_t$	—	absolute snapshot	yes
lag1	$\mathbf{x}_{t-1}$	own previous cycle	recent self	yes
diff1	$\mathbf{x}_t - \mathbf{x}_{t-1}$	own previous cycle	recent self	yes
roll5	$\mathbf{x}_t - \mathrm{mean}(\mathbf{x}_{t-4:t})$	own recent mean	recent self	yes
from_first	$\mathbf{x}_t - \mathbf{x}_1$	own origin (BOL)	origin self	yes
cycle (life fraction)	$2(t-1)/\max(T-1,1) - 1$	own entire lifetime	life fraction	no
cycle (fixed scale)	$(t-1)/200$	—	absolute position	yes

The **self-referenced lift (SRL)** makes the origin-self comparison explicit:

$$\tilde{\mathbf{x}}_t = [\,\mathbf{x}_t,\ c_t,\ \mathbf{x}_t - \mathbf{x}_1\,], \tag{1}$$

where c_t is the cycle-position column under the convention of the pipeline in use (Section 2.3) and $\mathbf{x}_1$ is the earliest retained record of the *same* cell—under Pipeline A the first row surviving the reference filter, which as Section 3.7 reports need not be the first recorded cycle, and under Pipeline B the first record whose inputs are valid. Throughout, the index t counts *retained records*, not rows of the released file: the reader of Section 2.3 drops rows whose inputs are unusable (and in the author protocol further rows by that protocol's own filter), and every family above—`lag1`, `diff1`, the five-record rolling mean and the cycle-position column—is defined on that retained sequence. In particular a "previous cycle" means the previous retained record, and the five-cycle window means five retained records. Two configurations are used: compact SRL (33 dimensions) for the pairing and anchor probes (Sections 3.9 and 3.11), and the full 81-dimensional stack, including all four history families, for the convention, ablation and published-reference analyses (Sections 3.1–3.3). The corrected-reader comparison evaluates both sets (Section 3.7); full-stack family importances are reported in Section 3.10. The external library retains its own 25 quality-checked features, giving a compact SRL set of 51 columns (Section 3.6). The companion deep-baseline comparison uses its 67-dimensional handcrafted features (Section 3.5), which the factorial analysis compares with the reference work's 16-dimensional set (Section 3.1). A 49-dimensional variant that appends a rolling five-cycle term was evaluated only as a slimming probe and was not adopted; it uses a rolling-median anchor (the median of the first ten cycles), not $\mathbf{x}_1$, and differs from the full stack by a median of $+0.4\%$ on gradient boosting. The compact set is not equivalent to the full stack on the frozen foundation model either: under the corrected reader of Section 3.7 the compact set's RMSE exceeds the full stack's by a

per-unit median of +13.9% (median of the per-unit relative changes, with the full stack as the denominator, the same aggregation the tables of that section use). Every feature obeys a *legality constraint* on their temporal transforms: they use only records $\leq t$ of the same cell. The cycle column is $c_t = (t-1)/200$ in Pipeline B and $c_t = 2(t-1)/\max(T-1, 1) - 1$ in Pipeline A, where T is the full retained length. Pipeline A's reader and normalization still use future information; the feature-transform test does not certify that pipeline as online-legal. Algorithm 1 lists the construction as implemented. The constraint is enforced by unit tests that perturb future rows and assert that all earlier feature rows are unchanged.

Algorithm 1. SRL construction for one cell, as implemented

1. Read the anchor once per cell: $\mathbf{x}_1$ is the 16-dimensional charge-segment statistic vector of the earliest retained record, and it is cached with the cell's identity.
2. For every retained record $t = 1, \ldots, T$, emit $\tilde{\mathbf{x}}_t = [\, \mathbf{x}_t,\ c_t,\ \mathbf{x}_t - \mathbf{x}_1 \,]$, where c_t is the cycle column of the pipeline in use and indices count retained records (Section 2.3).
3. In service the construction is streaming: row t uses only cycles $\leq t$, and $\mathbf{x}_1$ is read from the retained commissioning record rather than recomputed.

Legality test (shipped as a unit test and run on every build). For a random cycle index t, copy the cell, replace every row after t by noise, rebuild the history transforms on fixed input features, and assert that all rows up to t are bit-identical to the unperturbed build; any dependency on future rows fails the build.

Two optional components were evaluated and are not used by default. A five-cycle median anchor, intended for libraries whose first cycle is noisy, replaces $\mathbf{x}_1$ with the median of the first five cycles; it was worse than the fixed first cycle on 9 of 11 units (a pre-registered criterion asking for no degradation on at least 7 of 11 was not met), and it changed the sign of only one unit (HUST, +7.1% → −4.2%), so the fixed anchor is retained. A recovery-ratio gate applies monotone post-processing only when the source cells show a capacity-recovery ratio below 0.15; its precondition was verified on the independent library, whose source-side ratio is 0.0750, but it acts on the offline convention rather than on the deployment-computable pipeline and is therefore not part of the configuration reported here.

2.3. Two Evaluation Pipelines

Because the central issue is what can be computed in service, all experiments are run in one of two clearly separated pipelines, and numbers from the two are never pooled.

Pipeline A (reference convention, offline). Normalization follows the reference protocol: each cell is min–max scaled by its own full-lifetime extrema, and the cycle column is min–max scaled by the cell's full retained length. Its author-offline reader also filters rows using full-lifetime statistics including the capacity label (Section 3.7). This pipeline is used for like-for-like comparison against published results, since both sides then operate under identical rules.

Pipeline B (deployment-computable). Normalization statistics are fitted on source cells only and applied unchanged to held-out cells; the cycle column is scaled by a fixed constant ($(t-1)/200$) so that it does not depend on total life; all other features remain history-only. Each Pipeline B input is computable once the current cycle's required charge segment has completed; this is cycle-level availability, not continuous estimation during a partial cycle.

2.4. Models and Computational Setup

Two model families are used as-is, without architecture search:

- **GBDT**: histogram-based gradient boosting with library default hyper-parameters, as implemented by scikit-learn's `HistGradientBoostingRegressor`; five random seeds unless stated otherwise.
- **TabPFN**: a frozen tabular foundation model [14] used for training-free inference, with its in-context sample budget self-limited to 2048 rows; five seeds for the main comparison, three where stated. When a unit's training rows exceed that budget, the context is a uniform random sample without replacement, drawn per seed from the pooled training rows.

A tuned multilayer perceptron (L2 grid selected on source-side validation) is included as a third family for coverage analysis. The context budget was frozen before the comparisons of this paper and is not tuned per unit; a ladder over 2048, 8192 and the full training set (six units, two feature sets, three seeds) puts the median absolute change in per-cell RMSE at 5.8%, below the noise floor of Section 2.5, with the direction inconsistent (6 of 24 cells worsen). Four cells do exceed the floor, all of them the two TJU units with the history stack, so a larger budget helps where a unit has many training rows; we keep 2048 because it is protocol-frozen and because inference is 3–5 times cheaper. The subsampling rule is fixed and unit-independent: ten of the eleven units trigger it, and only XJTU-RW (1114 training rows) never does. The pools range from those 1114 rows to 107,850 for HUST. Because the sample is redrawn per seed, the seed-to-seed variance reported for a subsampled unit mixes model randomness with context sampling, and the ladder does not separate the two: a probe that crosses model seed with context seed at a fixed budget has not been run, so we report the mixing rather than an upper bound. The three-arm comparison of Section 3.9 draws the same context rows in every arm at every seed—the subsampler depends only on the row count and the seed, and the row count is identical across arms—so the pairing contrast is not driven by the context draw; this is a property of the implementation (checked in Section 3.9), not an assumption. Timings are reported as three separate classes and never pooled. *Archived runtimes* are whatever each result file records: for the foundation-model archive the "runtime" includes context preparation, its denominator includes training rows, and the hardware is not identifiable, so we derive no per-cell or per-unit figure from them. *This revision's measurement* (`bexp61`) was taken on one machine under the released environment (Python 3.14.6, NumPy 2.5.3 and scikit-learn 1.9.1): macOS arm64, 10 cores, the thread pools reporting `threads=10`, and the per-configuration load average in the result file. It covers the 17-, 33- and 81-dimensional stacks, a 2048-row context on two timing units (XJTU-2C and MIT), 24 repetitions at a batch of one and 24 at a batch of 1024: fitting takes 0.6–0.8 s per evaluation unit; a single-row prediction takes 2.8–4.3 ms (median; p95 $\leq$ 12.3 ms, $n = 24$ per configuration); serving a batch of 1024 costs 3.7–5.1 µs per row (medians; the slowest configuration's p95 is 5.7 µs); the process peak RSS is 181–328 MiB across the three stacks; and building one cell's full feature matrix costs 0.14–1.52 µs per retained row (amortized over the matrix; no in-service single-row update was measured). The *frozen foundation-model arm of Section 3.7* was run in this revision on CPU with the released v3 weights; each of its evaluation runs (one unit $\times$ view $\times$ seed, including context preparation and the predictions for that unit's held-out rows) took 1.3–60 s, with the device recorded in every record. The archived foundation-model replication of Section 3.10 was run on CUDA, also recorded per record. For reference, the physics-informed baseline trains repeatedly per unit (up to 200 epochs $\times$ ten repetitions) on its own hardware, and the deep baselines of the companion benchmark take about 170 s per run $\times$ five repetitions on theirs.

2.5. Metrics and Effect-Size Reference

The primary metric is root-mean-square error (RMSE) computed per held-out cell and then averaged across cells, which we verified to be the metric implemented in the reference

work's own scoring code. Published opponent values are taken from the officially released per-experiment results (ten repetitions per unit); we additionally executed the reference training code in our environment on one representative unit (XJTU-2C) and confirmed that the released values are consistent with code behaviour (ten-run distributions statistically indistinguishable, Mann–Whitney $p = 0.273$; our rerun mean is 9.4% higher, i.e., favourable to the reference).

Effect sizes are interpreted against a **historical threshold of 11.3%**. The value was fixed early in the study as a working scale and is reused unchanged; its original derivation is not documented, so we describe it as a heuristic of this study rather than as a measured property. A post hoc reconstruction from archived per-seed values is retained in the research archive: for each of 20 (unit $\times$ context-sampling-mode) groups on the history feature set, three seeds were run with nothing else changed, and the mean relative range $(\max - \min)/\text{mean}$ of per-cell RMSE across those groups is 11.35%. The agreement is consistent with the threshold's origin but does not establish it. Single-run differences below the threshold are not treated as findings. The threshold is a per-seed quantity: paired designs and twenty-seed means resolve smaller differences, and win counts are sign counts, so a 1–2% entry can decide a unit's sign without itself being claimed as an effect.

For the main comparison we test our per-seed values against the ten published per-experiment values with two-sided Mann–Whitney U tests, apply Holm correction across the 11 units within each backbone, and report bootstrap 95% confidence intervals of the relative difference. The corrected family is therefore composed of units, not of held-out cells, and a unit's weight in it is not proportional to its statistical power: the underlying test is against the ten published per-experiment values regardless of how many cells the unit holds out, and held-out counts range from two to twenty-three (Appendix A). Five seeds per unit permit, but do not guarantee, a significant result against the ten published values: with 5×10 observations the smallest attainable two-sided p is 6.7×10^{-4}, which can survive Holm correction across eleven units. Observed effects include small changes below the 11.3% descriptive noise floor (including significant GBDT improvements of 5.5% and 6.2%); that floor is not a significance threshold. Where an effect is of the order of the floor, as in Sections 3.9 and 3.11, we use twenty seeds instead.

2.6. Randomization-Based Validation Design

Feature-level claims are easily confounded, so the mechanism is tested with a three-arm randomized design in the tradition of randomization inference [24,25].

- **Treatment arm**: the true origin-self offset, $\mathbf{x}_t - \mathbf{x}_1$.
- **Mispairing arm**: the anchor is replaced by the first cycle of a randomly assigned *other* cell, using a fixed-seed derangement (no fixed points). The multiset of anchors is therefore unchanged, as are the arithmetic and the dimensionality of the transform: the same first cycles are reassigned. What changes is which origin each cycle is compared against, and with it the joint distribution of the resulting difference. This arm rules out the alternative that any difference feature would serve, whatever it is referenced to.
- **Instrument arm**: the anchor is the mean first cycle of the source cells, a per-column constant whose effect should be near zero for translation-invariant tree splits with matched randomness; finite implementations need not be exactly invariant. This is a negative control on the measurement apparatus itself [26]: its measured value tests the harness rather than the method, since a non-zero reading would mean the pipeline manufactures differences of its own.

For permutation inference we repeat the mispairing arm under M independent derangements and compute the Monte Carlo randomization p-value $p = (1 + \#\{m : T_m \leq T_{\text{self}}\})/(M+1)$, where T is the unit-averaged RMSE. We use $M = 100$ for GBDT

across all 11 units, and $M = 20$ for TabPFN on three representative units (chosen to span both carrier regimes of Section 3.10), reusing the identical derangements so that the two families are directly comparable. These are sampled randomization probabilities with the plus-one correction, not exhaustive enumeration of all possible anchor assignments; the minimum attainable values are $1/101$ and $1/21$.

Feature attribution uses permutation importance on held-out cells, aggregated by family, to describe fitted-model reliance on the feature families.

What the design does and does not establish. The manipulation acts on the feature matrix, not on the ageing process: the arithmetic and the dimensionality of the difference feature are held fixed and the anchor values are held fixed as a multiset, so what is broken is only the pairing between a cycle and the origin it is referenced to. The result is a randomization test of a representation claim, in the sense of permutation inference [24,25], and it supports statements of the form "the self-pairing, and not generic differencing, carries the gain". It does not identify a mechanism of electrochemical degradation, it does not estimate an interventional effect on a cell, and Pearl-style identification and do-calculus are outside its scope; the instrument arm is a negative control on the measurement harness rather than on the physical system [26]. Statements elsewhere in this paper about physical behaviour are mechanistic interpretations of that randomization result, not independent causal claims.

2.7. Pre-Registration and Reproducibility

Following pre-registration practice [27], 24 of the original 44 experiments carry 34 decision criteria in their frozen docstrings; this is not a claim that every experiment was pre-registered. Their documents are retained in the archive. The experiments added during this revision also state their criteria in the script docstring, and we grade their evidence rather than pooling it. For the scripts written first (the strict-reader re-run, the cost measurement, the curve-feature extraction and the two simple-baseline experiments), the commit carrying the criteria precedes the commit carrying the results, so the criteria are verifiable in the version history. One script family is weaker evidence: the TabPFN re-run on the second device shares its commit with its result, because it re-executes an already-frozen protocol on different hardware rather than introducing a new criterion. We therefore claim prior registration only for the former group. Other revision analyses whose script and first result share a commit (including bexp50–55) have explicit criteria but no separately timestamped pre-run commit; we do not claim prior registration for them. All outcomes, including falsified hypotheses, are retained in the research archive. Implementation corrections made during this revision carry per-script version notes rather than new criteria: the row-mapping rewrite and the CC/CV window correction of the raw-data manifest and curve extraction, the corrected MIT unit convention, the identity-gate exclusion in the baseline experiments, and the warm-up fix in the cost measurement; superseded archives are retained next to their replacements. Two of the notes record post-diagnostic rule corrections rather than pure implementation fixes—the cycle-duration quality gate and the identity-coverage threshold were added after inspecting the mapping diagnostics—and are marked as such in the scripts. Thirteen criteria relevant to this manuscript—those frozen in the previous revision, i.e., before the implementation corrections above—were not met, and the corresponding claims were downgraded or retracted rather than quietly re-worded. From the original study: the convention–importance correlation (Section 3.1), the five-seed win count (Section 3.9), the learned-anchor improvement and its autonomy variant (Section 3.11), the two distillation criteria and the injection criterion (Section 4.5). From this revision: two criteria that required the archived importance table to be reproduced within ten percentage points, by the ten-seed tree arm and by the foundation-model

arm (Section 3.10); two that required the corrected deployable pipeline to show an accuracy gain for the compact set and for the full stack under gradient boosting (Section 3.7); the corresponding criterion under the frozen foundation model (Section 3.7); and the fleet-cost criterion under its primary amortised reading (Section 3.7).

3. Results

3.1. The Normalization Convention Is a First-Order Variable

We first isolate the effect of the comparison target with a single-column manipulation. Holding data, model, splits, and all other columns fixed, we replace the life-fraction cycle column with the deployment-computable fixed-scale column (GBDT, five seeds, all 11 units). Table 2 reports the outcome and Figure 1 visualizes it.

Table 2. Effect of replacing the life-fraction cycle column with a deployment-computable column. All other factors are held fixed under the author-offline reader and per-cell normalization. RMSE pools test-cycle rows within each unit before averaging over five seeds; it is not equal-cell macro RMSE. Positive Δ means the deployable convention increases error. The last column gives the permutation importance of the life-fraction column for that unit (Section 3.10; ten-seed mean).

Evaluation Unit	Life-Fraction (RMSE)	Deployable (RMSE)	Δ (%)	Life-Fraction Imp. (%)
HUST	0.00944	0.01961	+107.8	95.6
TJU-3	0.00586	0.01011	+72.6	81.1
XJTU-R2.5	0.00804	0.00955	+18.8	79.2
XJTU-2C	0.00512	0.00570	+11.4	33.9
XJTU-satellite	0.00807	0.00850	+5.3	34.3
MIT	0.00762	0.00758	−0.5	5.7
TJU-1	0.01162	0.01076	−7.4	30.3
XJTU-3C	0.01172	0.01085	−7.4	75.7
XJTU-R3	0.00796	0.00692	−13.1	22.0
XJTU-RW	0.01117	0.00960	−14.1	93.8
TJU-2	0.01143	0.00577	−49.6	1.2

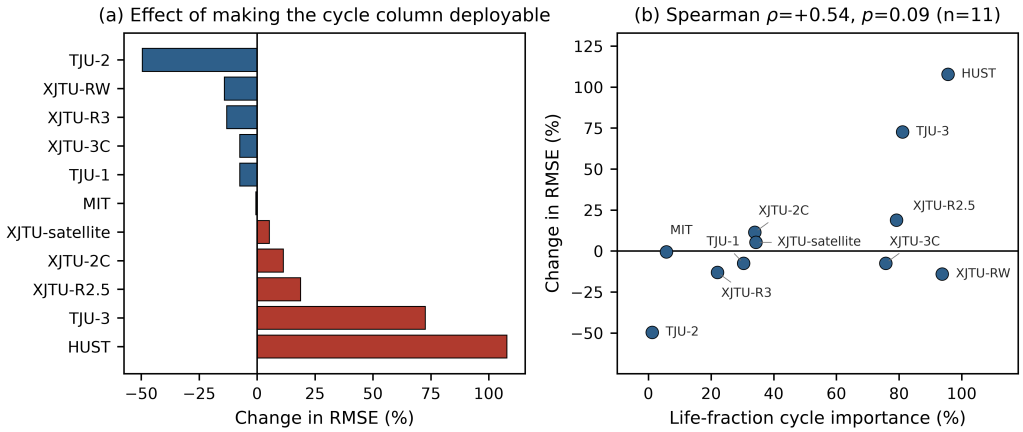


Figure 1. Effect of the cycle-column convention across all 11 evaluation units. (**a**) The same single-column manipulation moves error from −49.6% to +107.8%. (**b**) Sensitivity plotted against the permutation importance of the life-fraction column; the trend is consistent with the carrier account of Section 3.10 but is not statistically significant at this sample size (Spearman $\rho = +0.54$, $p = 0.09$, $n = 11$); at this sample size the confidence interval around ρ is wide, so the trend should not be over-read.

Two features of Table 2 matter. The magnitude is large: a single-column convention changes error by tens of percent on several units. The companion benchmark's architecture

spread is a separate MSE comparison under different features and splits, so its numerical range is not a calibrated measure of this effect's relative importance. The direction is not fixed: replacing the column helps TJU-2 (−49.6%), while it raises error on HUST and TJU-3. Those two units have high life-fraction importance, but XJTU-RW ranks second at 93.8% and improves with replacement. The positive rank correlation across 11 units ($\rho = +0.54$, $p = 0.09$) is therefore descriptive, not a predictive rule.

A second manipulation confirms that the convention governs the *fate of features* and not only accuracy. In a 2 × 2 factorial over feature richness (16-dimensional reference features versus 67-dimensional benchmark features) and normalization (per-cell versus source-fitted), the benefit of adding the explicit origin-self column reverses sign, as shown in Table 3. This probe was run under the author-preprocessing reader, whose row filter uses each cell's full-lifetime statistics; we therefore cite it as a legacy diagnostic about the normalization convention and not as evidence about the corrected deployable pipeline of Section 3.7 (the 67-dimensional cells follow the companion benchmark's own feature path, which has its own protocol).

Table 3. Median SRL gain relative to the corresponding baseline within each cell of a 2 × 2 factorial (GBDT, five seeds, XJTU, official held-out cells; author-preprocessing reader, i.e., a legacy diagnostic rather than the corrected deployable pipeline of Section 3.7). Negative values indicate improvement.

Feature Set	Normalization	Median SRL Gain (%)	Batches Improved
16-dimensional	per-cell (offline convention)	−6.7	6/6
16-dimensional	source-fitted statistics	−1.0	3/6
67-dimensional	per-cell (offline convention)	−5.3	4/6
67-dimensional	source-fitted statistics	+4.5	3/6

3.2. Which Comparison Target Carries the Benefit

An ablation over the four history families isolates the effective one. Table 4 reports median changes across the 11 units (GBDT, 550 runs).

Table 4. Ablation of history feature families relative to raw features (GBDT, author-offline reader, author normalization; negative is better). Each seed is scored by overall RMSE over all test-cycle rows pooled within a unit, then averaged over five seeds; relative changes are summarized by their median across 11 units. Longer cell trajectories therefore receive more weight than shorter ones. The equal-cell macro counterpart of this ablation is reported in Section 3.4; the two are different scoring experiments, not interchangeable estimates of the same contrast.

Configuration	Median Δ RMSE (%)
from_first (origin self)	**−4.4** (up to −32.6 on TJU-1)
+ lag1 (recent self)	−1.8
+ roll5 (recent self)	−1.9
+ diff1 (recent self)	−0.1
Full 81-dimensional stack	−10.9
Full stack minus from_first	−2.2
Full stack minus diff1	−11.0

Bold highlights the origin-self-only configuration; it does not denote the lowest RMSE change.

The metric difference reconciles Table 4 and the unit-level comparison in Section 3.4: their offset medians are −4.4% (pooled-cycle) and −5.6% (equal-cell), full-stack medians −10.9% and −7.2%, and TJU-1 offset changes −32.6% and −22.1%. Both use five-seed means within units, but the scoring weights differ. The equal-cell ladder of Section 3.4

additionally uses seed medians within units and is labelled separately. We retain both archived experiments and do not attribute their difference to seed aggregation alone.

Recent-self comparisons contribute little and the per-cycle difference is dispensable, whereas removing the origin-self family collapses the benefit from −10.9% to −2.2%. Two targeted manipulations test the dependence on the earliest retained own-cell record under the legacy protocol. Discarding each cell's first ten cycles, so that the anchor becomes the earliest *available* cycle rather than the original retained record, eliminates the benefit (median −5.8% → +2.2%; full curve in Section 3.11). Replacing the anchor with another cell's first cycle is not merely less useful but often harmful (Section 3.9).

3.3. Comparison with a Physics-Informed Network Under Its Own Protocol

We now enter the reference protocol of [1] without modification (its splits, its features, its normalization convention (Pipeline A), and its metric) and compare against its officially published per-experiment results. Table 5 reports the TabPFN backbone; Figure 2 shows both backbones.

Table 5. TabPFN with the full 81-dimensional history stack versus the published physics-informed results, under the reference protocol. Raw is the same backbone without the history families. Δ is the relative difference in per-cell RMSE; the 95% interval is a two-sample bootstrap over the per-seed runs reported here and the ten published per-experiment values of the reference (5000 resamples), so it describes run-to-run variability under the fixed official split and the fixed held-out cells, and is *not* an interval for a population of new cells; p values are Holm-corrected across the 11 units. Asterisks mark units where our result is significantly better. Seven of the eleven units hold out only two or three cells (the six XJTU batches and TJU-3); they enter the comparison as units, exactly as in the published protocol, and no cell-level inference is offered for them.

Evaluation Unit	Raw	This Work	Ref. [1]	Δ (%)	95% CI (%)	p (Holm)
XJTU-2C	0.00452	0.00425	0.00941	−54.8	[−60.0, −49.2]	0.0073 *
XJTU-3C	0.01317	0.01034	0.01218	−15.2	[−19.5, −10.6]	0.0073 *
XJTU-R2.5	0.00930	0.00832	0.01000	−16.8	[−25.7, −7.3]	0.0240 *
XJTU-R3	0.00935	0.00703	0.01051	−33.1	[−39.3, −24.7]	0.0073 *
XJTU-RW	0.01442	0.01196	0.01351	−11.5	[−16.8, −5.8]	0.0559
XJTU-satellite	0.00900	0.00569	0.00967	−41.1	[−44.8, −36.1]	0.0073 *
TJU-1	0.02050	0.01113	0.01579	−29.5	[−33.4, −25.2]	0.0073 *
TJU-2	0.00970	0.01319	0.01315	+0.3	[−9.6, +8.9]	0.9530
TJU-3	0.00513	0.00525	0.00789	−33.5	[−38.7, −27.7]	0.0073 *
HUST	0.00768	0.00781	0.00869	−10.1	[−14.6, −6.0]	0.0073 *
MIT	0.00743	0.00629	0.00740	−15.0	[−16.9, −13.0]	0.0073 *

Asterisks mark Holm-corrected $p < 0.05$; p values are from two-sided Mann–Whitney U tests comparing our per-seed results with the published per-experiment results.

The Raw column compares the same backbone without any history families against the full 81-dimensional stack; it does not isolate the offset alone. The raw backbones already sit near the published reference (median −7.0% for TabPFN and −4.7% for GBDT, lower on 7 and 6 of 11 units, respectively), and the full stack is lower on 10 of 11 for each backbone. For TabPFN, raw features outperform the full stack on HUST and TJU-3, consistent with, but not explained uniquely by, their high life-fraction importance. On TJU-2, TabPFN raw is 26% below the reference (0.00970 versus 0.01315), whereas the full stack reaches 0.01319. The separate GBDT attribution assigns that unit 77.9% offset importance, below MIT's 79.2% (Section 3.10); this is cross-model context, not attribution of the TabPFN contrast. Section 3.4 isolates the compact offset within GBDT.

The GBDT backbone reaches the same count: 9 of 11 units significantly better (−45.7, −17.4, −21.1, −24.3, −25.7, −19.1, −26.0, −5.5 and −6.2% on the significant units), with

TJU-1 (−5.6%) and TJU-2 (+2.7%) not significant. Across both model families the summary is therefore **9/11 significantly better, 2/11 not significantly different, and 0/11 significantly worse** for *each* backbone; the two non-significant units are not the same two from one backbone to the other (XJTU-RW and TJU-2 for the foundation model, TJU-1 and TJU-2 for gradient boosting), so neither is a single shared exception. Aggregated by library for the TabPFN backbone, the relative improvements are XJTU −27.1%, TJU −19.7%, HUST −10.1%, and MIT −15.0%; each aggregate is $100 \times [\sum_u \overline{\mathrm{RMSE}}_u^{\mathrm{ours}} / \sum_u \overline{\mathrm{RMSE}}_u^{\mathrm{ref}} - 1]$ taken over the units of that library, which is a ratio of summed means and therefore differs from averaging the per-unit percentages.

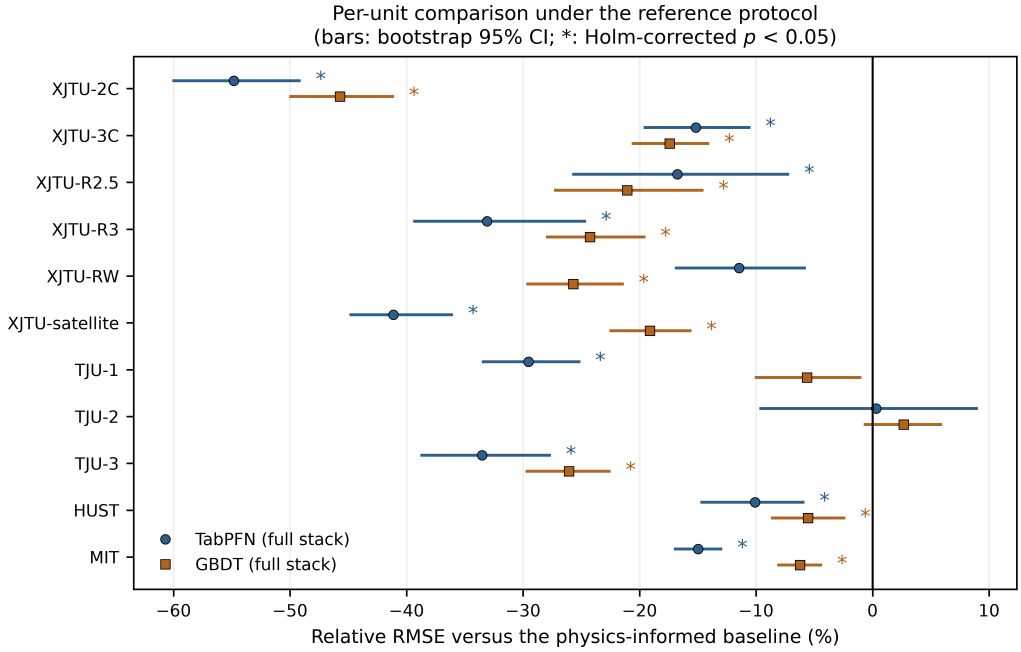


Figure 2. Per-unit relative error against the published physics-informed results for both backbones, each using the full 81-dimensional history stack. Bars are bootstrap 95% confidence intervals; asterisks mark Holm-corrected $p < 0.05$. Twenty of the twenty-two intervals lie entirely below zero.

3.4. How Much of the Margin Is the Representation?

The comparison of Section 3.3 changes several things at once, so we separate them here: Table 6 presents two further rungs of the baseline ladder that distinguish the SRL columns' contribution from the learner's, and two deliberately simple references place the machine-learning numbers in a familiar frame.

Under the author protocol the explicit offset does not help a linear model (median +4.1%, better on 4 of 11 units) but does help the tree model (−5.6%, 9 of 11); under the corrected deployable reader it helps neither (+0.8% and 5 of 11 for ridge, +4.3% and 5 of 11 for gradient boosting, from the same archive that Section 3.7 reports in full). The earlier positive entries came from the superseded preprocessing of Section 3.7 and are no longer used. The two learners still differ more than the two feature sets: with identical SRL features, gradient boosting beats ridge by about 36% under the author protocol and 25% under the corrected reader (relative to the ridge value), so the learner choice matters a great deal. We do not, however, turn that gap into a share of the margin against the physics-informed network, because the two comparisons differ in protocol, training budget and seed aggregation, and a margin measured under one protocol cannot be decomposed additively using a gap measured under another. What the tables support is the weaker statement that the learner and the feature set both contribute, and that the alternative

reading—one feature column making a simple learner beat a physics-informed network—is not what the data show.

Table 7 resolves the same question at unit level, within one backbone and one protocol, which the reference comparison of Section 3.3 cannot do on its own.

Table 6. Baseline ladder. Ridge regression (library default regularisation) and the gradient-boosted model on the same features, in both pipelines. The aggregation is two-stage: the median over five seeds within each unit, then the median across the eleven units; Δ is the median of the per-unit relative changes, so it is not the ratio of the two medians. "SRL" is the compact 33-dimensional self-referenced feature set; Δ is the relative change in per-cell RMSE from adding it to the corresponding raw baseline of the same row group, so a negative entry means the explicit offset helps. Rows labelled A are read under the author protocol, whose cycle column is the per-cell full-lifetime min–max convention of the reference work; rows labelled B are read under the corrected deployable reader of Section 3.7, whose cycle column is the fixed-scale one, and they are the same archive that Section 3.7 reports in full.

Pipeline	Learner	Median RMSE, Raw	Median RMSE, SRL	Δ (%), Units Improved
A (per-cell extrema)	Ridge	0.01203	0.01292	+4.1, 4/11
A (per-cell extrema)	GBDT	0.00915	0.00825	−5.6, 9/11
B (source-fitted, corrected reader)	Ridge	0.01167	0.01398	+0.8, 5/11
B (source-fitted, corrected reader)	GBDT	0.00918	0.01044	+4.3, 5/11

Table 7. Unit-level decomposition of the margin under one backbone and one protocol. Gradient boosting, Pipeline A, library defaults, 2048-row context, five seeds; each "ours" column is the mean of that unit's five per-seed per-cell macro RMSE values, so the three columns differ only in the feature set handed to the same learner, and the published column is the mean of the ten published per-experiment values for the unit. "Raw" is the 17-dimensional stack: the 16 cycle statistics plus the per-cell full-lifetime min–max cycle column of the author protocol, which is the cycle convention of every Pipeline A result here. "+SRL" adds the 16 offset columns (compact 33-dimensional set) and "+history" instead adds the four history families (full 81-dimensional stack). Δ_1 compares our raw arm with the published reference, Δ_2 the compact set with our raw arm and Δ_3 the full stack with our raw arm; the three are separate contrasts against different baselines and are not shares of a single margin. Negative is better. The archived runs of the frozen foundation model hold only the raw and 81-dimensional stacks, so its compact rung is missing here rather than interpolated.

Evaluation Unit	Published Ref.	Our Raw	Our +SRL	Our +History	Δ_1 (%)	Δ_2 (%)	Δ_3 (%)
XJTU-2C	0.00941	0.00524	0.00520	0.00511	−44.3	−0.6	−2.5
XJTU-3C	0.01218	0.01238	0.01029	0.01006	+1.6	−16.8	−18.7
XJTU-R2.5	0.01000	0.01025	0.00883	0.00789	+2.5	−13.8	−23.0
XJTU-R3	0.01051	0.00843	0.00818	0.00796	−19.8	−2.9	−5.6
XJTU-RW	0.01351	0.01082	0.01021	0.01004	−19.9	−5.6	−7.2
XJTU-satellite	0.00967	0.00835	0.00790	0.00782	−13.6	−5.3	−6.4
TJU-1	0.01579	0.01930	0.01504	0.01490	+22.2	−22.1	−22.8
TJU-2	0.01315	0.01540	0.01448	0.01350	+17.1	−6.0	−12.3
TJU-3	0.00789	0.00530	0.00601	0.00584	−32.9	+13.5	+10.2
HUST	0.00869	0.00827	0.00823	0.00820	−4.7	−0.5	−0.9
MIT	0.00740	0.00908	0.00695	0.00694	+22.8	−23.4	−23.6

The learner on raw features already sits below the published reference on 6 of 11 units (median −4.7%), and above it on TJU-1, TJU-2 and MIT (+22.2%, +17.1%, +22.8%), so it is not a uniform advantage. Adding the 16 offset columns to that learner helps on 10 of 11 units in sign terms (median −5.6%), as does the full history stack (median −7.2%); only TJU-3 worsens in sign, and HUST improves by an amount well below the noise floor of

Section 2.5 (−0.5% and −0.9%), so these are small descriptive changes rather than material gains. This pattern is consistent with high life-fraction reliance on those two units, but the counterexamples of Section 4.1 prevent extending it to a cross-unit rule. What the table does not license is adding Δ_2 and Δ_3 into one share of the headline margin of Section 3.3: those increments are measured against our own raw arm rather than the reference, the published column comes from a different training budget and scoring set, and the gap between our learner and the reference is itself unit-dependent.

The ladder of Section 3.4 is complemented by the simple references of Section 3.8, rebuilt from the raw curve releases under a protocol fixed before any model was fitted. The Coulomb- and fade-slope proxies reported in the first submission have been retired from the evidence: their input columns could not be traced to a verifiable definition in the released preprocessing code, so their numbers (a median per-cell RMSE of 0.113 and 0.115) are no longer used to rank against the learned models or to interpret why self-references work; they remain in the research archive as a historical record only. On the Severson comparison specifically, the landmark feature is the log variance of $\Delta Q(V)$ over an early cycle window; the processed releases used here retain no voltage curves, but the raw releases do—Section 3.8 reports the classic curve scalar adapted to the SOH task (the original early-life task itself remains unreproduced), rather than a proxy built from the released columns.

3.5. Comparison with Five Published Deep Baselines

Entering the companion benchmark's protocol [2] (leave-one-cell-out splits, 67-dimensional handcrafted features, its own normalization) we compare against the five deep baselines using its published appendix values (Table 8).

Table 8. Companion benchmark protocol: published deep baselines (MSE $\times 10^3$, three-run means) versus our results. "Fixed" is one configuration chosen in advance and applied to every batch (the frozen foundation model on the benchmark's own features); "Best of four" selects per batch among two feature sets × two backbones, so it is a per-batch choice among four configurations and is therefore more optimistic than the single fixed configuration tabulated beside it; the ratio is the best baseline divided by the best-of-four column as tabulated. The MSE $\times 10^3$ scale is the companion benchmark's own and is not comparable with the per-cell RMSE of Table 5 and Section 3.7: no conversion is shown, and rankings need not be invariant between the two metrics. Batches 1 and 4 differ by less than the per-run spread and are not individually decisive.

Batch	CNN	LSTM	GRU	MLP	Attention	Best Base.	Fixed	Best of Four	Ratio
1	0.060	0.039	0.028	0.034	0.078	0.028	0.044	**0.024**	1.17×
2	0.294	0.135	0.109	0.104	0.171	0.104	**0.034**	**0.034**	3.06×
3	0.042	0.039	0.026	0.024	0.058	0.024	**0.019**	**0.019**	1.26×
4	0.045	0.057	0.045	0.038	0.065	0.038	**0.037**	**0.037**	1.03×
5	0.237	0.172	0.154	0.451	0.180	0.154	0.096	**0.094**	1.64×
6	0.171	0.248	0.218	0.214	0.227	0.171	**0.164**	**0.164**	1.04×

Bold marks the best value in each row.

Selecting per batch among four configurations improves on the best of the five published baselines on all six batches; the single fixed configuration, which involves no per-batch choice, does so on five of six and loses only on batch 1. Because this protocol's per-cell normalization already supplies the self-comparison implicitly (Section 3.1), the margin here demonstrates the sufficiency of off-the-shelf carriers rather than an increment attributable to the explicit lift.

3.6. External Verification on an Independent Library

Table 9 reports an independent library excluded from feature design and model selection. Its raw statistics plus own-cell offset provide a separate pairing diagnostic under the legacy reader stated in the caption. The 100 sampled derangements characterize the randomized comparison, rather than a single mispairing draw. The row-selection path is not strict-online, so this is not external deployment validation.

Table 9. Independent library (45 cells): three normalization conventions × three arms (GBDT). "Mispaired" is the median of 100 independent derangements, each replacing the anchor with another cell's first cycle; p is the Monte Carlo randomization p value of the true pairing against that null. "Deterministic" here means a single run of the gradient-boosted model on all available training rows, with no context subsampling and no seed averaging; RMSE pools all held-out cycle rows, unlike the five-seed default used elsewhere; the derangement schedule is likewise fixed. The MATR-CLO legacy reader first screens cycles by label ($0.3 < Q_t/Q_{\mathrm{nom}} < 1.3$), selects columns using a full-trajectory cross-cell availability check, and drops remaining rows with missing inputs; all three normalization arms share this row set. Source-fitted normalization alone does not make this a strict-online reader result.

Normalization	Raw	SRL	Mispaired	SRL Gain (%)	p (100 Perm.)
per-cell full lifetime	0.02256	**0.01751**	0.02474	−22.4	0.0099
source-fitted statistics	0.01779	**0.01219**	0.01728	−31.5	0.0099
source-fitted + fixed-scale cycle	0.01806	**0.01217**	0.01715	−32.6	0.0099

Bold marks the lowest RMSE among Raw, SRL and Mispaired in each row.

The true pairing beats all 100 mispaired draws under each of the three normalization conventions, so on this library the self-pairing requirement holds whether or not the offline convention is in place.

3.7. Deployment-Computable Pipeline and Computational Cost

Table 10 reports Pipeline B re-run with a corrected reader that cannot see the target's future. The earlier implementation of both pipelines filtered rows using each cell's full-lifetime statistics *including its capacity label*, and its anchor was the first row surviving that filter rather than the first recorded cycle; measured on the released files, those anchors fall on records 1 to 4, and the filter changes the row set as well. Because that procedure uses target-future and label information, it cannot support a deployment claim. The corrected reader keeps a row only when the current row's inputs are finite, so nothing about the target's future enters either the row set or the anchor; the source-fitted normalization and the fixed-scale cycle column are unchanged.

In this corrected configuration the explicit reference families do not reach the pre-registered bar for a uniform gain, and their direction depends on the model and the feature set. Relative to raw features of the same backbone, the compact SRL set changes the median per-cell RMSE by +4.3% and improves 5 of 11 units for gradient boosting, and by +10.4% and 3 of 11 for the frozen foundation model; the full 81-dimensional stack gives +0.3% and 5 of 11 for gradient boosting and −1.2% and 6 of 11 for the foundation model; ridge shows the same pattern as the tree model (+0.8%, 5 of 11). Against a pre-registered criterion requiring improvement on at least 7 of 11 units, every arm fails, and we report the result as it stands. Direction is not uniform across models either: the foundation model pays substantially for the compact set while gaining slightly from the full stack. On XJTU-3C the compact set inflates that model's error by +154% (0.00780 to 0.01980) while the same columns move the tree model by +8%—the largest regression in Table 10—so the in-context learner is markedly more sensitive to the added offset block than the tree ensemble is. The earlier numbers are retained as a historical record and are not used. The corrected result

supports less than the earlier text claimed: the features are computable in service and the offline comparisons of Sections 3.3–3.5 stand under the reference protocol, but we no longer claim an accuracy benefit from the self-referenced lift under deployment conditions. The computational profile differs by backbone: a library-default gradient-boosted model fits in under one second per evaluation unit and predicts in milliseconds, whereas the frozen foundation model's CPU runs in this revision took 1.3–60 s per evaluation run (context preparation included, device recorded per record). The lift is also not uniformly harmless in this pipeline, as the XJTU-3C regression above shows. On cost we report only what was measured, under a stated scenario: a cell arriving with one computable current-cycle feature, a model already fitted, context already prepared, and the feature state updated incrementally. Note that "one cycle feature is available" is not the same as "a complete charge segment is available at any moment", which the scenario therefore assumes rather than derives. Under it, a gradient-boosted model fitted once on the 2048-row context costs 0.6–0.8 s per evaluation unit, a single-row prediction 2.8–4.3 ms at a batch of one, batched serving 3.7–5.1 µs per row at a batch of 1024, and the process peak RSS 181–328 MiB over the 17/33/81-dimensional stacks (Section 2.4); the measuring machine, its thread count and the library versions are recorded in the archived result file, and the models and environment in Section 2.4. Multiplying the measured model-prediction medians by 10,000 rows gives 28–43 s at a batch of one, or 0.037–0.051 s in batches of 1024; these arithmetic scenarios exclude fitting, feature acquisition and construction, scheduling and data transfer, and are not measured fleet runtimes. We report these as compute times only: no energy or carbon figure is derived from them, and no per-cell figure for the archived foundation-model runs (Section 2.4). The companion benchmark's deep baselines take about 170 s per run on their own hardware. Embedded feasibility is therefore left unverified: no measurement on microcontroller-class hardware was performed, and a pretrained transformer of this size is not in that class, so we claim neither embedded nor fleet-scale throughput. Section 3.8 reports the two validated simple references that a deployment-minded reader would try first.

Table 10. Pipeline B under the corrected reader: a row is kept only when its own inputs are finite, so no target-future or label information enters the row set or the anchor. Source-fitted normalization, fixed-scale cycle column, five seeds; medians of per-cell RMSE. "G" is the gradient-boosted backbone, "R" ridge and "T" the frozen foundation model (CPU, released v3 weights); "srl" is the compact 33-dimensional set and "hist" the full 81-dimensional stack. Bold marks the best configuration per unit. The values of the earlier implementation are kept in the research archive as a historical record and are not comparable.

Evaluation Unit	G-Raw	G-SRL	G-Hist	R-Raw	R-SRL	T-Raw	T-SRL	T-Hist
XJTU-2C	0.00918	0.00835	**0.00830**	0.01332	0.01982	0.00930	0.01027	0.01089
XJTU-3C	0.00864	0.00934	0.00914	0.01080	0.00929	**0.00780**	0.01980	0.01448
XJTU-R2.5	**0.00841**	0.01442	0.01239	0.01128	0.02146	0.01251	0.01217	0.01422
XJTU-R3	0.01343	**0.01177**	0.01356	0.01516	0.02434	0.01552	0.01608	0.01379
XJTU-RW	0.00664	0.00703	0.00654	0.00974	0.00989	**0.00607**	0.00832	0.00731
XJTU-satellite	0.01224	0.01281	0.01228	0.01514	0.01242	0.01021	0.01555	**0.00926**
TJU-1	0.00922	0.01044	0.01019	0.01781	0.01398	**0.00655**	0.00983	0.00839
TJU-2	0.00473	0.00434	0.00436	0.00666	0.00671	0.00415	**0.00363**	0.00387
TJU-3	**0.01119**	0.01167	0.01130	0.01167	0.01549	0.01262	0.01361	0.01247
HUST	0.02172	0.02071	**0.02059**	0.02485	0.02409	0.02374	0.02323	0.02141
MIT	0.00383	0.00380	0.00373	0.00569	0.00545	0.00204	0.00231	**0.00178**

3.8. Validated Simple Baselines on Raw Curves

Beyond the released-feature ladder of Section 3.4, a deployment-minded reader would try two simpler references. We rebuilt both from the raw releases under a protocol fixed before any model was fitted: official splits unchanged, deployable reader, the five learner arms of Table 11 scored on the same rows (retained-record index ≥ 100), GBDT at default settings; the physical reference reported below is scored on a subset of cells (one further requirement, stated below) and is therefore not placed in the same table as evidence of superiority. The first is the curve-difference scalar used for early life prediction on the MIT library [9], the base-ten logarithm of the variance of $\Delta Q(V)$ between the hundredth and the tenth discharge curves; it is computable after the first hundred cycles and therefore deployable. Its inputs are the XJTU archive and the MIT structs, which we obtained and aligned to the released cells (55 and 125 cells). The alignment is explicit per row: for 152 of the 180 cells the released discharge-capacity series reproduces a subsequence of the raw cycles to within 10^{-6} relative tolerance. Exclusion works in three layers, each stated where it applies: cells failing the identity gate are dropped from both the source and the target side of every learner arm; a target cell additionally needs its scored rows to pass the cycle-availability check (the curve scalar is only usable once the hundredth raw cycle has completed); and the physical reference below needs every scored row to map one-to-one onto a raw cycle, which is the strictest of the three. Failures at every layer are listed in the archived manifest.

Table 11. Simple curve-based references on the raw releases, scored on the rows with retained-record index ≥ 100 under the deployable reader; all values are cell-macro RMSE (medians over five seeds), lower is better. "raw", "+SRL" and "+history" are the released-feature learners on *this* scoring set and are not comparable with the full-life rows of Tables 5 and 10. Coverage is four of eleven units: the three TJU units and HUST have no raw curves in this study; the XJTU satellite batch and the R3/RW batches fail the identity gate (their released capacity series could not be verified against the raw cycles) and are excluded from every arm; within MIT one further held-out cell is excluded by the cycle-availability check.

Evaluation Unit	Cycle Only	Cycle + Scalar	Raw	+SRL	+History
XJTU-2C	0.01015	0.00645	0.00991	0.00917	0.00902
XJTU-3C	0.03411	0.03802	0.00992	0.01113	0.01079
XJTU-R2.5	0.01798	0.02511	0.00872	0.01640	0.01243
MIT	0.02837	0.02288	0.00384	0.00405	0.00378
Cross-unit median	0.02317	0.02400	0.00931	0.01015	0.00990

The scalar helps on two of the four units and hurts on the other two (median change $+0.0001$ RMSE), so we claim no uniform gain. The pre-registered improvement bar (7 of 11 units) could not have been met at this coverage, so we report the observation and the coverage limit rather than a failed test; nor is the ordering exceptionless, because on XJTU-2C the cycle+scalar arm (0.00645) sits *below* all three released-feature learners (0.00902–0.00991) on the same rows. On the remaining units the sixteen released statistics do carry information a cycle index plus one curve scalar does not (Table 11). The second reference is physical: the charge current integrated over the charge segment of the same cycle, divided by the nominal capacity, with one source-side efficiency calibrated on the first hundred cycles of the training cells (the label is discharge-side, so this is not a re-derivation of it). On the units whose rows map one-to-one onto raw cycles it reaches a cross-unit median cell-macro RMSE of 0.0053: on the three XJTU units 0.0009–0.0081 with $\eta \approx 0.999$–1.000, and on MIT 0.020 (cell median 0.004) with $\eta = 1.001$, where four cells are excluded by a cycle-duration data-quality gate. We therefore report it as a physical

consistency check rather than as an independent predictor, with its coverage (four of eleven units) stated.

3.9. Self-Pairing Is the Active Ingredient

Table 12 reports the three-arm randomized design of Section 2.6. The GBDT comparisons in this section pool all held-out cycle rows within each unit before seed averaging; the pooled randomization statistic then weights the 11 units equally. These metrics differ from the equal-cell macro RMSE used for the primary baseline comparison.

Table 12. Three-arm comparison, per-unit change relative to raw features (GBDT, twenty seeds; negative is better). The instrument arm is an affine negative control whose theoretical effect on tree models is zero. All arms use the author-offline reader and author-protocol normalization, including the full-lifetime, label-dependent row filter; these archived controls are offline diagnostics, not evidence for the corrected strict-online pipeline.

Evaluation Unit	Treatment: Self (%)	Mispaired (%)	Instrument (%)
XJTU-2C	−2.2	+3.4	0.0
XJTU-3C	−12.4	−13.2	0.0
XJTU-R2.5	−15.0	−13.3	0.0
XJTU-R3	−1.0	+4.0	0.0
XJTU-RW	−8.9	+15.4	0.0
XJTU-satellite	−3.4	+6.8	0.0
TJU-1	−33.7	−1.7	0.0
TJU-2	−3.1	+9.0	0.0
TJU-3	+12.6	+5.0	0.0
HUST	+8.1	+2.4	0.0
MIT	−23.5	+5.7	0.0

Three observations follow. First, the instrument arm is zero to within 0.04% on every unit across twenty seeds, as predicted by affine invariance, which confirms that the pipeline does not manufacture effects. Second, mispaired references are more often harmful than helpful: they raise error on 8 of 11 units (by up to +15.4% on XJTU-RW) and lower it on only 3, so the benefit is not generic trend information. On the frozen foundation model the contrast is larger still: on TJU-1 mispairing inflates error by a factor of 2.4 (0.00965 → 0.02330). Third, within single-condition batches a mispaired anchor necessarily comes from a cell operated under the *same* condition, and self-pairing still wins on 5 of 6 such units, which excludes condition matching as an alternative explanation.

The evidence in this section has distinct roles. The primary result is the pooled randomization test on gradient boosting: 100 sampled derangements, $p = 0.0099$. The instrument arm measures zero to within 0.04% on every unit, checking the harness under this implementation. Per-unit signs are descriptive and non-uniform: on TJU-3 self-pairing raises error by 12.6% against 5.0% for mispairing, and on HUST by 8.1% against 2.4%. High life-fraction reliance is consistent with these two observations, but does not predict the counterexamples in Section 4.1. XJTU-RW (self −8.9%, mispaired +15.4%) illustrates why arbitrary differencing is insufficient in this control. Section 3.10 describes model reliance; it does not establish information equivalence or predict SRL benefit.

Permutation inference quantifies this sampled comparison. Under 100 independent derangements the null distribution of unit-averaged RMSE has median 0.01107 and range $[0.01023, 0.01213]$; the true self-pairing attains 0.00916 and beats all 100 permutations, giving $p = 0.0099$ (Figure 3a). Repeating the procedure for the frozen foundation model with the identical derangements on three representative units yields 20/20 wins on TJU-1 and MIT and 18/20 on HUST, with a pooled permutation $p = 0.0476$ and a null median of 0.02006 against an observed 0.00898. That arm supports the same conclusion on a second

model family, though with $M = 20$ its p value cannot fall below 0.0476 by construction; the stronger of the two tests is the 100-permutation result on gradient boosting. Taken together, the three arms separate the pairing from everything else that the feature changes: in the mispaired arm the arithmetic and the dimensionality are identical and the anchors are only reassigned across cells, so the multiset of anchors is preserved and what is broken is the pairing between a cycle and its origin; the advantage goes with it.

Two pre-registered criteria in this analysis were not met and the corresponding claims were downgraded rather than removed. The rank correlation of Section 3.1 did not reach significance at $n = 11$, and extending the three-arm comparison from five to twenty seeds moves the per-unit win count from 9/11 to 8/11: XJTU-2C reverses to a win, while XJTU-3C and HUST reverse to losses; we report the twenty-seed value as the more reliable estimate. The two runs also resampled the mispairing permutation, so per-unit differences reflect both sources of variation; the estimate that does not depend on any single draw is the 100-permutation p value above. Directional outcomes on the large-effect units are unchanged.

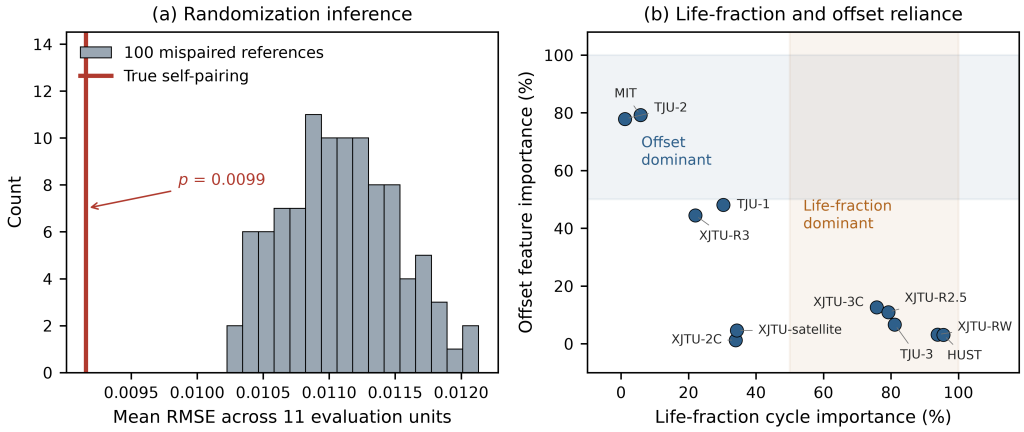


Figure 3. Pairing controls and model attribution. (**a**) Randomization inference: the true self-pairing (red line) against the null distribution of 100 mispaired references ($p = 0.0099$, gradient boosting). (**b**) Life-fraction and offset importance shares vary across units (ten-seed means of Section 3.10, gradient boosting, author-protocol normalization); shaded regions mark the offline-only and offset-dominant regimes; these regions do not imply information equivalence or predict accuracy.

3.10. Life-Fraction and Offset Reliance

Permutation importance, aggregated by family, describes reliance on the life-fraction and offset columns for the fitted tree model under author-protocol normalization. Their relative shares vary across units (the ten-seed replication below; Figure 3b). Table 13 retains the historical single-run importance analysis.

Because the archived analysis in Table 13 motivated the carrier account, and because the configuration that generated it could not be confirmed from the research archive, we recomputed the attribution with a documented protocol (gradient boosting, ten seeds, full 81-dimensional stack) and report that replication in Table 14; the importance statements in the text are taken from that replication, and the archived single-run values of Table 13 are retained as the historical record they are. The new tree analysis shows distinct family-reliance patterns, and the replication also quantifies what a single run could not: the per-unit standard deviations span 0.3 to 11.3 percentage points, and the largest difference from the single archived run is 15 pp, concentrated in the cycle/from_first pair, whose features can be correlated, without demonstrating information equivalence. We read that as an explanation of the discrepancy rather than as an identification of the archived protocol, and the replication's standard deviations describe the new ten-seed runs: they are not an error bar on the archived single values. We also asked whether the archived values came

from a different backbone, by replicating the same estimator with the frozen foundation model (two seeds, CUDA): that arm departs from the archived values by up to 62 pp (mean 11.8 pp), against 15 pp (mean 3.0 pp) for gradient boosting. Neither arm reproduces the archive within 10 pp, so its exact configuration remains unconfirmed; the foundation-model arm is, however, much farther from it and does not exhibit the two-carrier structure, so the archived table is better described as consistent with the tree model than as reproduced. The unit where this matters most is XJTU-R3, where the two families are 22.0 ± 4.2 (life-fraction) and 44.5 ± 11.3 (offset), so a single run can legitimately report them as close or as separated. We therefore report Table 13 as a description of a regime and the standard deviations of Table 14 as uncertainty of the new replication only, rather than treating either as a calibrated quantity.

Table 13. Permutation importance by feature family (%) on held-out cells, from one run of the archived importance analysis whose generating configuration could not be confirmed; it is retained as the historical record, and the importance statements in the text are taken from the ten-seed replication below. The life-fraction column exists only offline; the offset column is computable in service.

Evaluation Unit	Life-Fraction Cycle (%)	Origin-Self Offset (%)
HUST	**99.4**	0.4
XJTU-RW	**98.4**	0.6
TJU-3	**96.1**	0.6
XJTU-R2.5	**93.6**	3.3
XJTU-3C	**87.0**	4.5
XJTU-satellite	**35.8**	3.9
XJTU-2C	**35.3**	0.7
XJTU-R3	**34.2**	30.8
TJU-1	24.3	**46.2**
MIT	3.9	**89.3**
TJU-2	0.8	**86.6**

Bold marks the larger of the two importance shares in each row.

Table 14. Ten-seed replication of the permutation importance of Table 13, and the source of the importance statements in the text (mean $\pm$ standard deviation, percentage points; gradient boosting, author-protocol normalization, full 81-dimensional stack, 2048-row context budget; per-column permutation with five repeats, scored by per-cell macro RMSE, negative importances clipped at zero, then summed within family and normalized to 100%). Only the life-fraction and offset families are shown; "gap" is the difference of the means in favour of the leading family.

Evaluation Unit	Life-Fraction Cycle	Origin-Self Offset	Leading Family (Gap)
HUST	95.6 ± 0.3	3.0 ± 0.3	cycle (92.6)
XJTU-RW	93.8 ± 0.3	3.1 ± 0.3	cycle (90.7)
TJU-3	81.1 ± 1.7	6.7 ± 3.1	cycle (74.4)
XJTU-R2.5	79.2 ± 1.6	10.9 ± 1.7	cycle (68.3)
XJTU-3C	75.7 ± 2.1	12.6 ± 1.3	cycle (63.1)
TJU-2	1.2 ± 0.5	77.9 ± 3.6	offset (76.7)
MIT	5.7 ± 2.9	79.2 ± 3.1	offset (73.5)
TJU-1	30.3 ± 4.5	48.1 ± 4.6	offset (17.8)
XJTU-R3	22.0 ± 4.2	44.5 ± 11.3	offset (22.5)
XJTU-satellite	34.3 ± 3.2	4.6 ± 0.8	cycle (29.7)
XJTU-2C	33.9 ± 2.5	1.2 ± 0.4	cycle (32.7)

3.11. Which Early Cycle Serves as the Anchor

The mispairing control supports an advantage for own-cell anchors in the legacy aggregate comparison, with per-unit exceptions. To test whether the earliest retained

record is preferable within that protocol, we replaced it with a learned convex combination of each cell's first ten retained records, $\mathbf{r} = \sum_{i=1}^{10} \alpha_i \mathbf{x}_i$. An attention scorer learned the weights on training cells; the same gradient-boosted learner then evaluated both anchors. This isolates the tested anchor construction from predictor choice. Both probes of this subsection use the author-offline row set; a ten-record anchor also requires that window to have completed.

Accuracy does not improve, and with sufficient seeds the fixed choice is the better one. Under twenty seeds the learned anchor raises per-cell RMSE on 7 of 11 units, by a median of $+2.0\%$; the individual differences are small but consistent enough to resolve, and after Holm correction across units the learned anchor is significantly worse on six units and significantly better on one (Figure 4b). The six penalised units are the six XJTU batches, while the single improvement is -18.6% on TJU-2; the split does not track cell counts or cycle counts (Spearman $\rho = +0.22$, $p = 0.52$ against the shortest cell length), so we report it as a pattern without offering a mechanism. A five-seed version of the same comparison put the median at $+3.0\%$ and resolved nothing; the twenty-seed run was pre-registered with an explicit reversal criterion (a Holm-corrected significant penalty on at least four units forces the boundary-condition reading), which is what triggered the present account. Repeating the comparison with the frozen foundation model points the same way (median $+3.2\%$, learned anchor lower on 5 of 11 units), though that arm cannot establish per-unit significance: with three seeds per arm the smallest attainable two-sided rank-test p is 0.10, so no unit can survive Holm correction across eleven of them by construction.

The learned weights are only mildly tilted toward the earliest cycles (median weight 0.15 on the first cycle against 0.10 for a uniform allocation, median mass 0.42 on the first three cycles, and the largest weight falling on the first cycle in 7 of 11 units), and the resulting anchor departs substantially from the first cycle in every unit (median maximum deviation 2.81 in normalized feature units). Averaged over units the weight profile decreases monotonically across the window, from 0.16 on the first cycle to 0.07 on the tenth, crossing the uniform level near the fourth (Figure 4a): the preference for early cycles is systematic but not exclusive, and that residual spread is what costs accuracy.

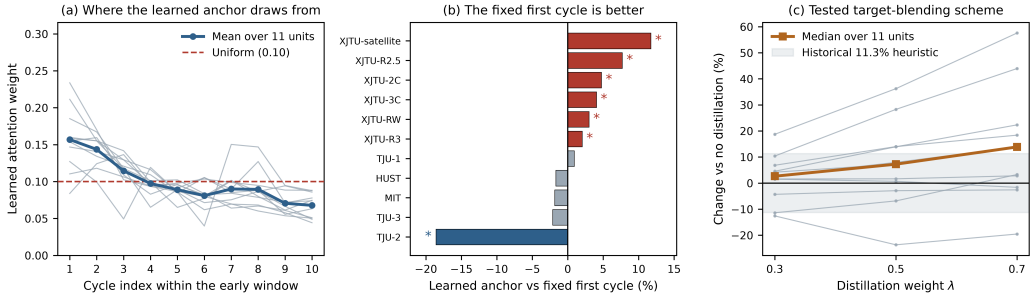


Figure 4. Two probes of the reference construction. (**a**) Learned attention weights over the early window; grey lines are the eleven evaluation units, the dashed line marks a uniform allocation. (**b**) Per-unit change in per-cell RMSE from replacing the fixed first cycle with the learned anchor (twenty seeds); red bars are significantly worse and the blue bar significantly better after Holm correction; grey bars indicate differences that are not statistically significant after Holm correction. Asterisks mark Holm-corrected $p < 0.05$. (**c**) Change in per-cell RMSE when predictions from a teacher using the offline-only carrier are blended into an online-computable student's target, against the blending weight λ; grey lines show individual evaluation units, and the orange line shows their median. The shaded band is the historical 11.3% heuristic, not a confidence interval.

Read together with Table 15, these probes favour the earliest retained own-cell record over the tested delayed and learned alternatives under the offline protocol. They do not establish a unique manufacturing state, exclude all other anchor constructions, or show that

the same advantage survives the corrected reader. Retaining early records is necessary to reproduce the tested SRL construction, but its online accuracy must be assessed separately (Section 3.7).

Table 15. Sensitivity to anchor availability (GBDT, three seeds): median SRL gain when the first K cycles of every cell are discarded. The probe was run under the author-preprocessing reader (offline row set), so it is read as a diagnostic of the offline comparison rather than as a statement about the corrected deployable pipeline of Section 3.7.

Cycles Discarded (K)	Median SRL Gain (%)	Units Improved
0	−5.8	8/11
10	+2.2	5/11
50	+4.9	3/11
100	+3.0	4/11

4. Discussion

4.1. Carrier Attribution: Scope and Counterexamples

The carrier pattern of Table 14 is consistent with HUST and TJU-3, where life-fraction importance is high and the explicit offset provides little benefit in the legacy controls. It does not predict the sign of the benefit across units. XJTU-RW has 93.8% life-fraction importance, yet replacing the cycle column reduces error by 14.1% (Table 2), and true self-pairing reduces error by 8.9% against a 15.4% increase when mispaired (Table 12). XJTU-3C and XJTU-R2.5 also improve despite high life-fraction shares (75.7% and 79.2%). Permutation importance describes reliance of a fitted model under one protocol; it neither establishes redundant information nor supplies a general rule for when SRL helps. We therefore restrict the carrier explanation to the observations it fits and report these counterexamples explicitly.

Deployment removes the life-fraction channel because total life is unknown while a cell is in service, whereas the own-cell offset remains computable from retained records. Whether that improves accuracy is a separate question. The corrected evaluation of Section 3.7 answers it negatively for the learners tested: removing target-future and label information from the reader does not meet the cross-unit improvement criterion for any explicit-reference arm. The offline attribution therefore does not establish an accuracy advantage after deployment.

The anchor experiments further delimit the legacy result: discarding the first ten retained cycles removes the median benefit under the tested offline protocol (Table 15). This supports retaining the earliest recorded cycle for that comparison; it does not identify a manufacturing state or establish a strict-online accuracy advantage.

Subtracting a cell's earliest record can remove a time-invariant component of its charge-segment statistics and retain changes associated with ageing. This is a physical interpretation of the representation, not an identity with the target. In the released primary benchmark and our adapter, SOH is cycle discharge capacity divided by the dataset's nominal capacity, Q_t / Q_{nom}, not by the measured capacity of that cell's earliest record. It therefore need not equal one at the anchor. A reference from another cell can introduce between-cell offsets; this interpretation is consistent with the legacy mispairing result but does not identify an electrochemical degradation mechanism or imply strict-online accuracy improvement.

BatLiNet [31] distinguishes intra-cell early-cycle differences from inter-cell references to predict battery lifetime across ageing conditions. Our task is cycle-level SOH estimation, and our legacy control reassigns own-cell anchors while holding their values fixed. Its result concerns that representation under its reader and learner, not the usefulness of

inter-cell learning on a different prediction task; it does not contradict BatLiNet's learned inter-cell reference.

4.2. Design Implications of the Negative Results

The negative results constrain three separate interpretations. First, the offline anchor probes favour the earliest retained record over the tested delayed and learned anchors; they do not show that later cycles cannot reconstruct a useful reference in general. Second, the tested teacher-target blending scheme does not meet its improvement criterion: the student improves on only 3 of 11 units and the median error increases with blending weight (Figure 4c). The mixed directions of the legacy cycle-convention contrast do not identify why blending fails because they concern a different manipulation and row protocol. Third, the corrected reader shows that compact SRL is not uniformly harmless: for TabPFN it raises error on XJTU-3C (Table 10). A deployment evaluation must therefore specify retained inputs, anchor availability and model-specific accuracy rather than assume the offline advantage transfers.

4.3. Comparability with Published Numbers

The XJTU dataset has an officially maintained summary of published results [32] spanning dozens of 2025–2026 papers, with reported RMSE values as low as 0.23%. That summary also records that evaluation *modes* differ across papers (intra-cell interpolation versus cross-cell estimation, single batch versus all batches, and non-uniform splits), and its maintainers do not rank across papers. The XJTU-family studies cited in the Introduction were not audited paper by paper in this revision: we did not verify, from their full texts or released code, that they adopt the official splits or the per-cell normalization of the release used here, so we place no numbers side by side with theirs, and the comparability statement below is restricted to results published under the reference protocol itself. We follow the same discipline: comparisons are made only within protocols aligned position by position, by reading the reference implementation's scoring code and by executing its training code. Our per-unit RMSE values (0.43–1.32%) are of the same magnitude as recent reports while being obtained entirely in the cross-cell mode under fixed official splits, which is the harder setting. One preprint [8] reports values below ours on the same four public libraries (387 cells), but it reports batch-level point values without run-to-run distributions, discloses neither its cell-level splits nor its code, and its chemistry description (four chemistries including LCO) is inconsistent with the official releases of these libraries (NCM, NCA and LFP; Appendix A); it is therefore cited without comparison. The follow-up studies cited in Section 1 are not uniformly comparable at this granularity: they do not all state whether the released cell-level splits are retained, and some report pooled rather than per-experiment error. We therefore restrict quantitative comparison to the protocol whose splits and scoring code we could verify directly [1].

4.4. Implications for Battery Management in Sustainable Power Systems

Three consequences concern potential deployment. First, source-trained default gradient boosting and frozen-model inference provide baselines that do not require target-cell fitting; under the corrected reader, adding SRL does not establish an overall accuracy advantage. Second, CPU cost was measured for separate workloads: fitting the gradient-boosted model takes under one second, its single-row prediction 2.8–4.3 ms, and the frozen foundation model takes 1.3–60 s per unit $\times$ view $\times$ seed including context preparation (Section 3.7). The sub-second fitting measurement belongs only to GBDT. Third, reproducing this SRL requires retaining the earliest valid charge-segment record with cell identity; doing so preserves feature availability, not a demonstrated online accuracy benefit. No microcontroller measurement or field deployment was performed. Prior deployment-oriented

work has examined data sufficiency and short observation windows [33]; our reader audit concerns a separate constraint, the information available at scoring time.

4.5. Limitations

The claims of this paper hold under five stated conditions, which distinguish protocol-specific results from deployment hypotheses.

Anchor availability. The implemented SRL requires an earliest retained own-cell record. Table 15 shows that discarding ten retained records removes the median benefit under the legacy offline reader. This does not prove that SRL is unusable for all second-life cells or fleets without commissioning data: a later observed anchor remains computable, but reproduces a different representation whose accuracy is unvalidated here. The corrected-online result must be distinguished from this offline sensitivity probe.

Model-family coverage. A tuned MLP improves on only 5 of 11 units; adding offsets to the companion benchmark's own MLP improves on only 1 of 6 batches. These observations do not identify why those networks fail to exploit the offsets. The positive full-stack comparisons for GBDT and TabPFN are offline and include learner effects; neither supports a universal model-family or deployment accuracy claim.

The tested transfer scheme did not meet its criterion. Motivated by the offline carrier pattern, an obvious remedy is to train on that channel and transfer what it knows to a deployable model. This is legitimate: complete trajectories of the *training* cells are available, so the constraint binds only at inference on the target cell (learning under privileged information [34]), and we tested it: a teacher using the life-fraction carrier supplied out-of-fold soft targets that were blended into an online-computable student's regression objective. The tested blending scheme did not meet its improvement criterion. The student improved on only 3 of 11 units, the median change was $+7.3\%$ at a blending weight of 0.5, and degradation grew monotonically with that weight ($+2.6\%$, $+7.3\%$, $+13.9\%$ at 0.3, 0.5 and 0.7, Figure 4c). Per-unit Mann–Whitney tests with Holm correction leave one unit significantly better and none significantly worse, so the accurate reading is that this scheme did not meet its improvement criterion; neither universal transfer failure nor significant harm is established. One pattern is consistent with Table 2: the offline cycle convention is not a universal upper bound: it is better than the fixed-scale convention on 5 of 11 units, but exceeds the 11.3% noise floor on only 4 (XJTU-2C, XJTU-R2.5, TJU-3 and HUST), and is worse on the other 6. These are legacy convention diagnostics, not corrected-online comparisons, and do not identify the reason target blending fails. Whether a different transfer mechanism succeeds where target blending fails is open, and identifying the failure mechanism would require a different probe.

Statistical power at the unit level. Cell-level generalization is limited by the two or three held-out cells in each XJTU batch and TJU-3. The legacy per-unit pairing effect-to-noise ratios span 0.5–11.9, and six of eleven units reach $p \leq 0.1$; the pooled randomization result concerns an average across units. These pairing tests differ from the run-level comparison against the published baseline: in the latter, two units per backbone are non-significant, with three distinct units in their union (TJU-2 for both, XJTU-RW for TabPFN, TJU-1 for GBDT). Non-significance does not establish equivalence.

Scope of the data and of the claim. All libraries are laboratory ageing data acquired under controlled conditions; field data with dynamic loads, temperature excursions, and calendar ageing is not covered. All claims are scoped to battery SOH estimation, where the benchmark target is cycle capacity divided by nominal capacity; we make no claim about degradation tasks whose labels lack that structure.

5. Conclusions

What a feature is compared against (other cells, a recent past, or the earliest retained record) is a design decision that has been made implicitly inside normalization choices, and it materially affects error under the tested legacy protocol: replacing a single column's convention moves error from -49.6% to $+107.8\%$ across eleven evaluation units. The benchmark SOH label uses nominal capacity as its denominator, while the prevailing per-cell convention depends on full-lifetime extrema that a cell in service cannot know. We made the comparison explicit and online-computable as a self-referenced lift.

Under the offline author protocol, off-the-shelf learners with the full history stack are significantly more accurate than the physics-informed network on 9 of 11 units for each of two backbones, with none significantly worse. Under the corrected strict-online reader, compact SRL instead changes median per-cell RMSE by $+4.3\%$ for gradient boosting (5 of 11 improve) and $+10.4\%$ for the frozen foundation model (3 of 11 improve); feature computability does not translate into an overall accuracy gain. The companion benchmark improves on five published deep baselines in 6 of 6 batches with per-batch selection (5 of 6 with a fixed configuration), but its normalization already supplies an implicit comparison. The unseen-library 22–33% improvement and the randomized pairing results ($p = 0.0099$ for gradient boosting, $p = 0.0476$ for the frozen foundation model) are legacy diagnostics under their stated readers. Tree-model carrier attribution is consistent with HUST and TJU-3 but has counterexamples and does not establish a universal mechanism.

For sustainable power and energy systems the practical message is economical rather than architectural. The measured CPU workloads have seconds-per-unit cost without training on the target system; the SRL feature contract additionally requires each cell's earliest record to be retained, with the cost profiles of the two backbones reported as separate measurements of different workloads and neither presented as an embedded-deployment claim. Three qualifications travel with that statement. The deployability result is negative in our corrected evaluation: with target-future and label information removed from the reader, the reference families do not meet the stated cross-unit improvement criterion, so what survives is the feature contract rather than an accuracy claim. Every accuracy figure is obtained on laboratory ageing data under controlled protocols, so what is established is that each required quantity is computable at time t in service, not that the same accuracy transfers to field BMS conditions with dynamic loads, calendar ageing and sensor drift; extending the analysis to such field data, and to fleet-level deployment where per-cell origin records and cross-fleet heterogeneity interact, is left to future work. The whole account also presumes a retained commissioning record: without it, the reference cannot be reconstructed after the fact. We recommend that the comparison target be reported as a standard ablation in degradation modelling, that evaluation protocols distinguish channels available offline from those that survive deployment, and that commissioning records be treated as a functional part of the health-estimation pipeline.

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Data Availability Statement: All battery datasets analysed in this study are third-party open-access releases and are available from their original publications: the four processed libraries and official splits from [1], the companion benchmark data from [2]. The independent verification library is the closed-loop-optimisation batch of [30], used in preprocessed form. The analysis code, feature

implementations, experiment scripts with their decision criteria, and the result archives supporting the reported analyses, including the scripts and results added during revision, are openly available at https://github.com/13927/srl-battery-soh (accessed on 3 October 2026).

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BMS	Battery management system
BOL	Beginning of life
CI	Confidence interval
GBDT	Gradient-boosted decision trees
LCO	Lithium cobalt oxide
LFP	Lithium iron phosphate
MLP	Multilayer perceptron
NCA	Lithium nickel cobalt aluminium oxide
NCM	Lithium nickel cobalt manganese oxide
RMSE	Root-mean-square error
SOH	State of health
SRL	Self-referenced lift

Appendix A. Dataset Characteristics

Table A1 summarizes the eleven evaluation units. Cell counts, cycle-count ranges and the lowest observed SOH are computed from the preprocessed release actually used here; chemistry and nominal capacity are the metadata of that release. Ageing conditions differ by library: the six XJTU batches are released as separate charge/discharge protocols (2C, 3C, R2.5, R3, a random-walk profile, and a satellite duty profile); the TJU cells span 25, 35 and 45 °C with charge rates of 0.25C–1C and discharge rates of 1C–4C, as encoded in their cell identifiers; the HUST cells share one charge protocol and differ in 77 discharge protocols [29]; the MIT cells come from a fast-charging protocol study [9]. All are laboratory ageing data, which is the scope stated in Section 4.5.

Table A1. The eleven evaluation units of the primary benchmark. Held-out cells follow the official splits without modification. Cycles per cell is the range across the cells of that unit; SOH_{min} is the lowest state of health observed in the unit.

Evaluation Unit	Cathode	Nominal Capacity (Ah)	Cells	Held Out	Cycles per Cell	SOH_{min}
XJTU-2C	NCM	2.0	8	2	355–379	0.798
XJTU-3C	NCM	2.0	15	3	116–307	0.808
XJTU-R2.5	NCM	2.0	8	2	490–641	0.775
XJTU-R3	NCM	2.0	8	2	459–633	0.797
XJTU-RW	NCM	2.0	8	2	132–266	0.801
XJTU-satellite	NCM	2.0	8	2	602–869	0.805
TJU-1	NCA	3.5	66	13	26–790	0.694
TJU-2	NCM	3.5	55	11	42–1246	0.654
TJU-3	NCM+NCA	2.5	9	2	857–964	0.659
HUST	LFP	1.1	77	20	1115–2672	0.800
MIT	LFP	1.1	125	23	34–2095	0.818

MATR-CLO (external verification library) is the closed-loop-optimisation batch of [30]: the same cell type, cycled at 30 °C, in which the charging protocol of each cell was selected by Bayesian optimisation from a candidate set of six-step, ten-minute fast-charging protocols, with a 4C discharge.

Table A2 collects these ageing conditions in one place. It reports what the original releases and their source publications state, and marks a parameter as not reported rather than inferring it. The six XJTU batches are released as separate charge/discharge protocols and we keep their names literally: 2C and 3C denote a 2.0C and a 3.0C charge followed by a 1.0C discharge, R2.5 and R3 denote variable-rate discharge to 2.5 V and 3.0 V, respectively, RW is a random-walk discharge profile, and satellite is a geostationary duty profile. The TJU cells are named CY{temperature}-{charge rate}_{discharge rate}, so the three batches span 25, 35 and 45 °C with charge rates of 0.25C–1C and discharge rates of 1C–4C. For HUST, the release states that all cells share one fast-charging protocol whose rate is given only in the source paper's supplementary information, while the 77 discharge protocols differ from cell to cell; the MIT unit is the fast-charging study of [9], whose cells were cycled at 30 °C with multi-step fast charging and a 4C discharge to 2.0 V.

Table A2. Ageing conditions of the libraries behind the eleven evaluation units. Values are taken from the original releases and their source publications; "n.r." marks a parameter the release does not state. Charge and discharge rates are given as C-rates relative to the nominal capacity of the cell.

Library (Units)	Cell	Cathode	Nominal (Ah)	Charge	Discharge	Temperature (°C)
XJTU (6)	LISHEN 18650	NCM	2.0	2.0C (2C, R2.5, R3, satellite); 3.0C (3C); random-walk profile (RW)	1.0C (2C, 3C); {0.5, 1, 2, 3, 5}C to 2.5 V (R2.5) or 3.0 V (R3); random walk (RW); 0.667C GEO duty (satellite)	room (n.r.)
TJU (3)	18650	NCA/NCM/ NCM+NCA	3.5/3.5/2.5	0.25–1C	1C (TJU-1, TJU-2); 1–4C (TJU-3)	25, 35, 45; 25 (TJU-3)
HUST (1)	A123 APR18650M1A	LFP	1.1	one common fast-charging protocol (rate n.r.)	77 cell-specific multi-stage profiles	30
MIT (1)	A123 APR18650M1A	LFP	1.1	multi-step fast charging, 3.6–6C mean to 80% SOC, then 1C CC–CV	4C to 2.0 V	30
MATR-CLO (external)	A123 APR18650M1A	LFP	1.1	closed-loop-optimised six-step 10-min fast charging	4C	30

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